

Early blight

Recognize the problem

During the rainy season plants may have dark brown rings (leaf spots), surrounded by a yellow ring. The spots start first on the oldest leaves. This disease is called early blight or alternaria and affects melongene, peppers, tomatoes and other plants in the potato family.

Background

This disease is caused by a fungus. The spores or seeds of the fungus are spread by wind or water splash. This disease gets worse in warm and wet conditions. These spores survive in infected plant residue left in the field and in soil for a long time.

Management

- Keep the crop dry by planting crops wide apart to allow free flow of air between plants. For example, plant tomato at least 60-90 centimetres (two to three feet) apart.
- Keep the plants healthy during the first 6 weeks after transplanting by using recommended fertilisers. Additionally, you can buy a fungicide at the agro-shop. Apply the fungicide at the rate shown on the label.
- Prune overgrown plants to keep humidity low and to let in the air. Remove diseased plant parts from the field and dispose. Do NOT plough debris into soil!
- Mulch around plants to reduce soil-water splash and help keep spores off healthy leaves.
- Do not replant these types of crops (peppers or tomatoes) in the same field for at least two to three years. This is called 'crop rotation'.
- Apply a copper-based fungicide to infected plants at the prescribed rate on the label every 14 to 21 days or as recommended.

Early blight on potato leaves.
(Photo by Farmers Guide)



Early blight on tomato leaves.
(Photo by Mike Rutherford, CABI)



When using a pesticide, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, and pre-harvest interval.

Scientific name(s) > *Alternaria solani*

The recommendations in this factsheet are relevant to: Trinidad and Tobago

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Edited by Plantwise.

Early Blight on Potato

Alternaria solani



Symptoms of early blight on potato. Photo: Howard F. Schwartz, Colorado State University, Bugwood.com



External and internal view of Early Blight lesions; note symptoms within this tuber are shallow (Sandra Jensen, Cornell University, Bugwood.org)

Prevention	Monitoring	Direct Control	Direct Control	Restrictions
- Plant tolerant varieties when available - Select site with good drainage and avoid fields bordered by trees and dense vegetation - Remove weeds to reduce prolonged moisture and high humidity around the plant - Eradicate potential sources of inoculum by removing plant residues after harvesting and volunteer plants - Practice crop rotation for 3 to 4 years with non-susceptible crops, such as leguminous crops (beans, cowpeas) or cereals - Good sanitation practices - wash field equipment and machinery after use - Plant at recommended spacing to allow for airflow for optimal relative humidity - Respect planting time seasons	- Inspect the field once a week - The first symptoms are small light to dark green, circular to irregular-shaped water-soaked spots - These lesions usually appear first on the lower leaves. Lesions often begin to develop near the leaf tips or edges, where dew is retained the longest - White mycelium develop on the underside of the leaf and on stems throughout the canopy. - Act immediately when first symptoms are noticed	- Cut off stems three weeks before harvesting in order to reduce infection of tubers - Bury infected plants and plant parts below ground at a depth of about 60 cm or more away from the production site	- Wear protective clothing when handling pesticides - Consult recent list of registered pesticides (Agri-input Plant Health Division) - Follow label instructions carefully - Dispose pesticide containers responsibly, Do not allow product to enter the waterways or sewer	WHO Class U (Unlikely to give acute hazard in normal use). PHI 7 days.



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Early blight on Tomato

Recognize the problem

Tomato plants infected with early blight develop small dark brown to black spots on lower shaded leaves, stems and fruits. Leaves develop spots that are brown and leathery with faint concentric rings that look like a target. The spots later become uneven in shape. Spots on fruits often occur near the calyx end, where the fruit attaches to the plant.

In the seedbed the small plants wilt and eventually die. In older crops, stem death occurs while leaves fall off the crop and fruits drop early. The fruits are then exposed to sunburn.

Background

Early blight is a seed-borne disease that results in poor germination of seedlings. Disease development is most serious during warm wet conditions. It is caused by a fungus that also attacks old transplanted seedlings and plants that are stressed due to poor fertilization or attack by nematodes. The disease is also spread by wind, rain, running water, farm tools and also farm workers.

Early blight on tomato leaf. (Photo by Kengap Horticulture)



Management

Non-chemical control:

- Use certified disease free seeds and tolerant varieties.
- Practice crop rotation with non-host plants and plants that do not belong to the tomato family.
- Avoid plant injury during transplanting.
- Ensure a strong plant growth so that plants can escape attack of the fungus.
- Plants should be staked and pruned correctly to allow good air flow and avoid disease spread.
- Avoid planting tomatoes next to related crops like eggplant, pepper and potatoes.
- Remove and destroy the old plants and weeds.

Chemical control:

- Some of the effective chemicals available include Mancozeb, Azoxystrobin, copper oxychloride, Cymoxanil + Famoxadone, Dimethomorph + Mancozeb.

When using a pesticide, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, and pre-harvest interval.

Scientific name(s) > *Alternaria solani*

The recommendations in this factsheet are relevant to: many countries where this pest or problem is present



Authors: This factsheet is based on information written for "The Tomato Farming Handbook", first published by KENGAP Horticulture, 2011.
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Late blight on Tomato

Recognize the problem

Late blight is a minor disease during the dry season; however it is most serious during cool moist conditions. Spores from the sick plant can move into a clean plant and infect it.

At first, the disease causes irregular greenish or water soaked lesions on the leaves, stems and fruits. Leaf spots turn a bluish-grey colour and sometimes when the weather is wet have a white fuzzy growth around the margins of the spots on the undersides of the leaves. Infected leaves turn brown and wither, but stay attached to the plant. Fruits develop watery brownish green spots which develop on the upper half of fruit. The disease leads to a very quick death of the entire plant.

Background

The disease is spread by water splashes from the soil. The disease can also be spread by overhead irrigation and in overcrowded farms. The disease survives in old crop residues and other plants related to the tomato. There are varieties of tomatoes that are tolerant to the disease.

Management

Non-chemical control:

- Practice crop rotation by avoiding tomato or related crops (e.g. Irish potatoes, capsicum, brinjals etc) for at least 3 to 4 seasons.
- Plant resistant varieties.
- Transplants should be free of disease.
- Stake the plant to keep them off soil.
- Mulch the plant to reduce water splashes.
- Irrigating during the day can help reduce the severity of the infection because this disease needs to have free water around to spread.

Chemical control:

- Fungicide sprays should be applied as soon as the disease symptoms are seen, or before if there are prolonged periods of wet weather as these conditions favour the disease.
- These chemicals include copper oxychloride, Dimethomorph + Mancozeb, Famoxadone + Cymoxanil, Metalaxyl-M + Mancozeb and Mancozeb.

Late blight on tomato. (Photo by Phil Taylor, CABI)



Late blight on tomato leaves. (Photo by Phil Taylor, CABI)



When using a pesticide, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, and pre-harvest interval.

Scientific name(s) > *Phytophthora infestans*

The recommendations in this factsheet are relevant to: many countries where this pest or problem is present



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Potato Late Blight

Recognize the problem

This fungal disease affects leaves, stems, roots and tubers of potato. It is an important disease in Afghanistan as it causes high economic losses to farmers. The disease is identified by black or brown spots on the underside of leaves and stems. They are small in the beginning and appear wet. The disease spreads quickly, as the spots become larger and destroy the leaves. The whole plant can be destroyed within a few days. Diseased tubers are often copper brown, reddish or purplish in colour and feel soft to the touch.

Potato Late Blight Attack on Leaves. (Photo by Howard F. Schwartz, CC BY)



Background

During winter, the fungus generally survives underground in the tubers and in other host plants. The disease can be spread by wind so it can affect neighbouring fields very quickly. Cool and moist conditions also help to spread the disease. The fungus is washed off from the leaves, enters the soil and infects the potato tubers. The fungus remains in the tubers and can cause disease to affect potato in the next season.

Potato Late Blight on Tuber. (Photo by Scott Bauer, CC BY)



Management

- Use disease-free, healthy seeds
- Use resistant varieties
- Ensure plant to plant distance is 25cm in a row
- Plant in recommended time for sowing
- Keep plants healthy by using balanced fertilizers
- Collect all dry plants and burn
- Harvest only when plants are dead
- Avoid harvesting in wet conditions
- Store potatoes in good storage conditions with an air ventilation facility
- Spray with a fungicide like mancozeb (e.g. Dithane or manzate), metalaxl-M (e.g. Ridomil gold) and meneb with recommended doses during all the season
- Treat the seed before sowing with fungicide like mancozeb 80% w/w @ 4 g/litre

When using a pesticide, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, and pre-harvest interval.

Scientific name(s) > *Phytophthora infestans*

The recommendations in this factsheet are relevant to: Afghanistan



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Potato late blight

Recognize the problem

This common disease of potato can be devastating unless controlled. It affects leaves, stems and tubers, turning them brown or black and rotten. This disease will significantly reduce yield and make potato rot in store. Early blight is a similar disease, but early blight has rings inside the brown spots. Late blight does not have rings in the spots.

Background

Late blight is more common in wet weather. The disease infects leaves and will spread to the tubers. The disease can be carried in seed potatoes.

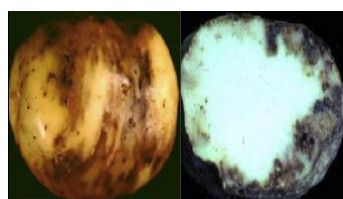
Management

- Inspect your crops often and act as soon as you see late blight in your potatoes.
- As soon as you see the disease, spray with Ridomil Gold (Metalaxyl and Mancozeb) at a rate of 35 ml per 20 ltr knapsack sprayer.
- Spray every two weeks if the weather is wet and every three weeks if the weather is dry until two weeks to harvest. Spray when the leaves are dry.
- Use clean seed potatoes, from a source you trust.
- After harvesting potatoes, plant a different crop the following year.
- Moulding soil along the potatoes to protect the tubers from the disease.
- Remove and destroy all stems and leaves at least two weeks before harvesting, so the disease does not spread from the leaves to the potatoes when harvesting.
- Some potato varieties are somewhat resistance to late blight such as Tigoni (Kenya) and Rutuku (Uganda).

Late blight symptoms on leaves.



Tuber infected with late blight.



When using a pesticide, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, and pre-harvest interval.

Scientific name(s) > *Phytophthora infestans*

The recommendations in this factsheet are relevant to: Ghana, Kenya

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